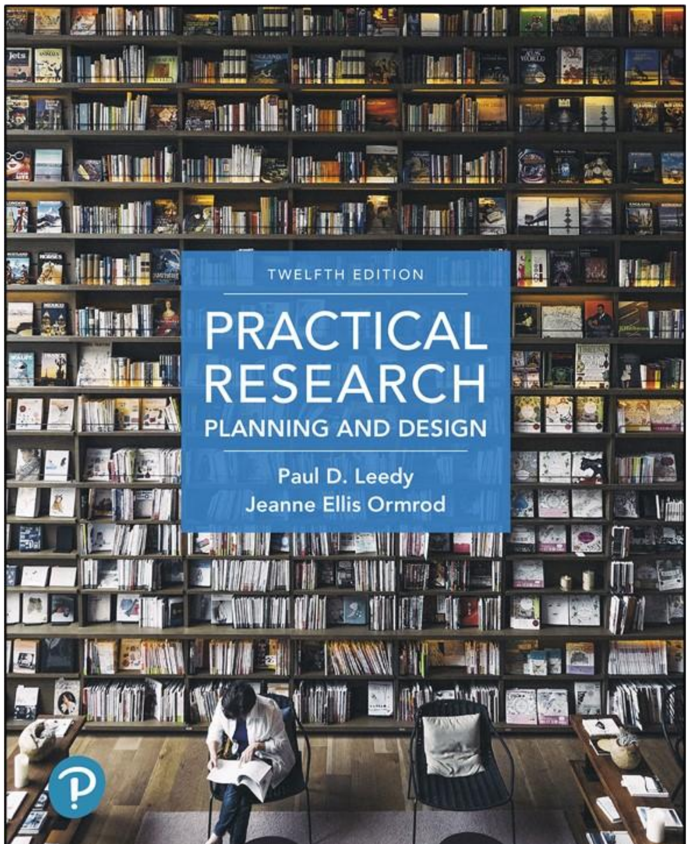
# Practical Research: Planning and Design

Twelfth Edition




## Chapter 11

Analyzing Quantitative Data

# Exploring and Organizing a Data Set

- Statistics: computational procedures that enable us to find patterns and meaning in numerical data
- Data need to be organized before they are analyzed
- Whatever the researcher does to prepare will affect the meaning that the data reveal
- Every researcher should be able to provide a clear, logical rationale for the procedure used data

# Organizing Data

- To identify patterns, first organize the data in meaningful ways
  - In a table
  - In a graph

# Use the Computer to Organize and Analyze Data

- Electronic spreadsheet: a software program that allows a researcher to manipulate data displayed in a table
- Uses of electronic spreadsheets:
  - Sorting data
  - Recoding data
  - Calculating formulas
  - Graphing from the data
  - Employing “trial and error” explorations

# Choosing Statistics

- Two major functions of statistics
  - **Descriptive statistics** describe what the data look like, how broadly they are spread, and how closely two or more variables within the data are associated with another.
  - **Inferential statistics** allow us to draw inferences about large populations by collecting data on relatively small samples.

## Statistics as Estimates of Population Parameters

- Parameter:
  - characteristic or quality of a population
    - Parameters of a circle
      - diameter is twice the radius (r)
      - circumference is always $2\pi r$
      - area is always $\pi r^2$
- Statistic
  - Any calculation we perform for the sample rather than the population

# The Nature of the Data (1 of 5)

- The nature of the data helps determine the statistic that is necessary
  - Have your data been collected for a single group or for two or more groups
  - Were continuous or discrete variables involved
    - Continuous = infinite number of possible values
    - Discrete = finite, small number of possible values that are independent of each other

# The Nature of the Data (2 of 5)

- The nature of the data helps determine the statistic that is necessary
  - Do you have nominal, ordinal, interval, or ratio data
    - Nominal = separate, non-ranked categories
    - Ordinal = sequenced categories
    - Interval = sequenced categories with equal units of measurement
    - Ratio = equal intervals AND true zero point

# The Nature of the Data (3 of 5)

- Many statistical procedures are built on the notion of a normal distribution. You may know this as the “bell curve” or the “normal curve.” This normal distribution has several distinguishing characteristics:

1. It is horizontally symmetrical.
2. Its highest point is at its midpoint.
3. Predictable percentages of the population like within any given portion of the curve.
   – 68-95-99.7 rule

# The Nature of the Data (4 of 5)

- The nature of the data helps determine the statistic that is necessary
  - Are the data normally distributed, skewed, or kurtic?
    - Normal: Horizontally symmetrical frequency distribution – scores at the center are most frequent
    - Predictable percentages of the population lie within any given portion of the normal curve

# The Nature of the Data (5 of 5)

- The nature of the data helps determine the statistic that is necessary
  - Skewed: Not symmetrical – scores that are most frequent can be anywhere along the X axis
    - Peak to the left of midpoint = positively skewed
    - Peak to the right of the midpoint = negatively skewed
  - Kurtic: peaked/pointy (leptokurtic) or flat (platykurtic)
    - Percentile ranks are an example

# Choosing between Parametric and Nonparametric Statistics

- Parametric statistics are used when
  - data reflect an interval or ratio scale
  - data fall in a normal distribution
- Nonparametric statistics are used when
  - data are ordinal in nature
  - data distribution is non-normal

# Descriptive Statistics

- Measures of central tendency
- Measures of variability
- Measures of association

# Measures of Central Tendency (1 of 2)

- Techniques for finding a central point around which the data revolve
- Descriptive statistics **describe** a body of data.
- Three things a researcher might want to know about the data set:
  1. Points of central tendency
  2. Amount of variability
  3. Extent to which two or more variables are associated with one another

# Measures of Central Tendency (2 of 2)

- Mode: the single number or score that occurs most frequently
- Median: the numerical center of a set of data – the number in the middle of the (ordered) set of scores
- Mean: the arithmetic average of the scores within the data set
  - May need geometric mean for some data sets (e.g., growth)

# Measures of Variability (1 of 3)

- Extent to which data points cluster around point of central tendency
  - Range: value reflecting the spread of data from lowest to highest value (highest score minus lowest score)
  - Interquartile range: Quartile 3 – Quartile 1
  - Five-number summary (lowest, highest, Q1, Q3, and median)

# Measures of Variability (2 of 3)

- Average deviation: the average of differences of each score and the mean score
- Standard deviation: standardized way of computing the distance between each score and the mean; takes negative numbers into account
  - the measure of variability most commonly used in statistical procedures

# Measures of Variability (3 of 3)

- Norm-referenced score: reflects where each person in the group is, relative to other members of the group
- Standard score: tells us how far an individual's performance is from the mean, with respect to standard deviation units

# Keep It in Perspective

- Statistics related to central tendency and variability provide a beginning point from which to view data
- Statistical manipulation of the data is not research
- Research demands interpretation of the data

# Measures of Association

- Correlation: statistical process that allows the researcher to identify any relationship between variables
  - Results in a statistic (number) between −1 and +1
  - This number is known as a correlation coefficient

# Correlation (1 of 2)

- Correlation tells us direction
  - Positive: as one variable increases, the other variable also increases
  - Negative: as one variable increases, the other variable decreases
- Closer to +1 or −1: strong correlation
  - Variables closely related
  - Can make predictions
- Closer to 0: weak correlation

Pearson

# Correlation (2 of 2)

- A variety of statistical tests measure correlation
- Correlation does not necessarily imply causation
  - Ask yourself, “What might be the underlying reason for the association?”

# Inferential Statistics

- Used to draw inferences about a population from a smaller sample
- Two main functions:
  1. Estimate population parameter from a random sample
  2. Test statistically based hypotheses

# Estimating Population Parameters (1 of 2)

- Point estimate: single statistic that is taken as a reasonable estimate of the corresponding population parameter
  - Example: Sample mean used to estimate population mean
  - Typically does not correspond exactly with its true equivalent in the population

# Estimating Population Parameters (2 of 2)

- Interval estimate: a range within which a population parameter probably lies
- Sometimes called "confidence interval"
  - Attaches a certain level of confidence that the estimated range does indeed contain the population parameter

# Testing Hypotheses (1 of 2)

- Researchers begin with a null hypothesis
  - Any result observed is the result of chance alone

# Testing Hypotheses (2 of 2)

- Testing the Null Hypothesis
  - Using data from a sample to make inferences about the population from which the sample is drawn
  - Are these results simply “due to chance” – the fact that this specific sample was tested
  - Do the results accurately reflect the population
- Researchers reject “chance alone” if the results surpass a pre-set cutoff level
- “Significance Level” or “alpha level” (a)

# Errors in Hypothesis Testing (1 of 2)

- Type I error
  - Incorrectly rejecting the null hypothesis
  - “This result is not due to chance” – but it really IS due to chance

# Errors in Hypothesis Testing (2 of 2)

- Type II Error
  - Concluding that a result was due to chance when in fact it was not
  - Failing to reject a null hypothesis that is actually false
  - Also known as a beta error

## Suggestions for Increasing the Power of a Statistical Test

- Use as large a sample as is reasonably possible
- Maximize the validity and reliability of your measures
- If logically defensible and logistically practical, obtain repeated measures of your dependent variable
- Use parametric rather than non-parametric statistics whenever possible

# Meta-Analysis

- Used to analyze and draw conclusions about other researchers' statistical analyses
- When conducting a meta-analysis, the researcher:
  - Conducts an extensive search for relevant studies
  - Identifies appropriate studies to include in the meta-analysis
  - Converts each study's results to a common statistical index

# Statistical Software Packages

- Commercially available options include:
  - SPSS, SAS, SYSTAT, Minitab, Statistica
- Freeware programs include AdaMSoft and R, for example
- Advantages of using a statistcal program
  - Range of available statistics
  - User-friendliness
  - Assumption testing
  - Graphics

# Interpretation of the Data

- Interpreting the data means several things:
  1. Relating the findings to the original research problem and to the specific research questions and hypotheses
  2. Relating the findings to preexisting literature, concepts, theories, and research studies
  3. Determining whether the findings have practical significance as well as statistical significance
  4. Identifying limitations of the study

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